

Global and Local Projection-Based Reduced-Order Models

2D Burgers and Hypersonic Vehicle Campaigns

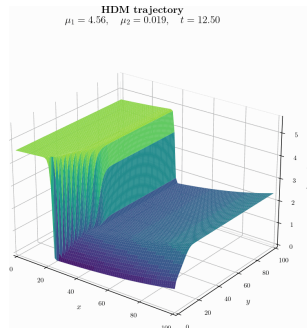
Sebastian Ares de Parga¹, Radek Tezaur¹, Charbel Farhat^{1,2}

¹Department of Aeronautics and Astronautics

²Institute for Computational and Mathematical Engineering
Stanford University, Stanford, CA 94305, USA

September 21, 2026

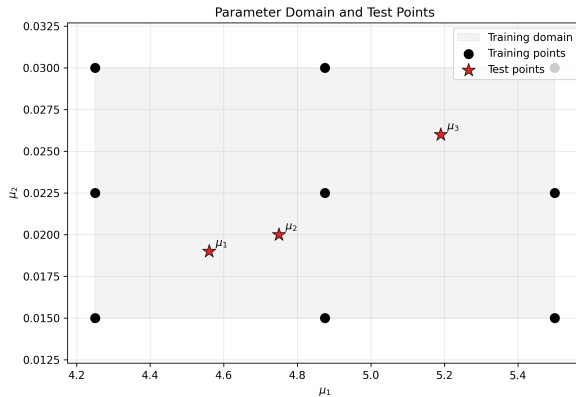
2D Burgers Campaign



Global and local HPRM validation

Linear POD bases, quadratic manifolds, and GPR manifolds

Parameter Domain and Test Points



Online Performance (250×250): Avg/Max Error over 3 Test Points

Computational model	n (N for HDM)	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,u}$ (%)	max $\mathbb{RE}_{2,u}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	–	12.517	–
HPROM	96	–	4 241	1.030	1.096	0.366	34.2
HQPROM	39	–	3 460	0.818	0.861	0.641	19.5
HPROM-GPR	20	131	1 799	0.666	0.785	0.236	53.1
Local HPROM ($N_c = 3$)	35–48	–	3 405	0.897	0.933	0.154	81.4
Local HQPROM ($N_c = 3$)	21–27	–	2 152	0.888	1.025	0.215	58.3
Local HPROM-GPR ($N_c = 3$)	9	60–97	811	0.714	0.786	0.113	110.5
Local HPROM ($N_c = 5$)	23–31	–	2 384	0.915	0.943	0.095	132.0
Local HQPROM ($N_c = 5$)	15–20	–	1 593	0.760	0.878	0.128	97.9
Local HPROM-GPR ($N_c = 5$)	7	41–68	631	0.673	0.880	0.094	133.2

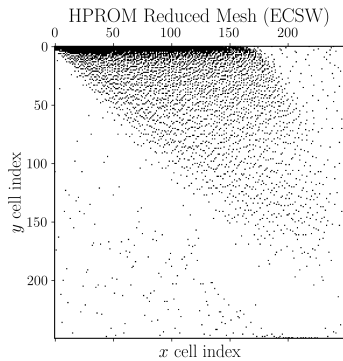
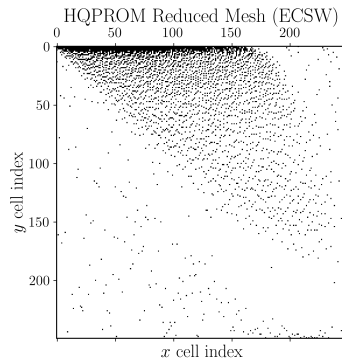
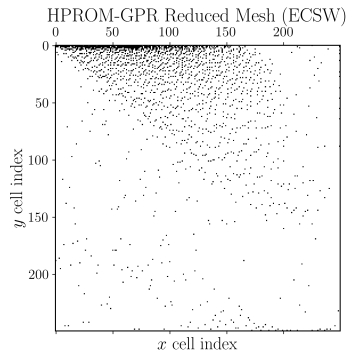
Speedup is computed with respect to the HDM runtime, $t_{\text{HDM}} = 751.043$ s, averaged over the three test points.

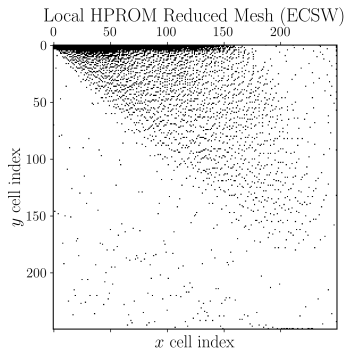
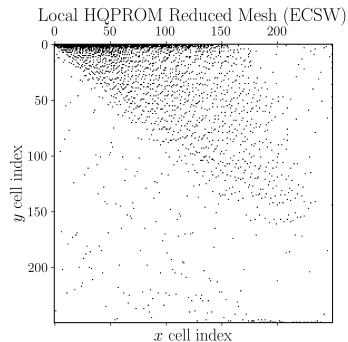
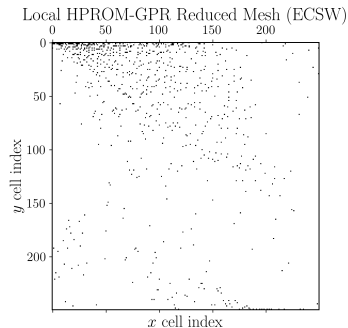
Online timings were measured on Sherlock using one node, 24 MPI ranks, and 24 cores.

N_e denotes the number of sampled ECSW elements ($N_e = 62,500$ for HDM/full mesh).

All models are reported at a matched accuracy target of $1\% \pm 0.2\%$ in max $\mathbb{RE}_{2,u}$: for each N_c the accuracy knob is tuned into that window (pod.to1 for Local HPROM, ζ_{qua} for Local HQPROM, n for Local HPROM-GPR), so differences in n , N_e and speedup are read at constant accuracy.

Global Reduced Meshes (ECSW)

HPROM ($N_e = 4,241$)HQPROM ($N_e = 3,460$)HPROM-GPR ($N_e = 1,799$)

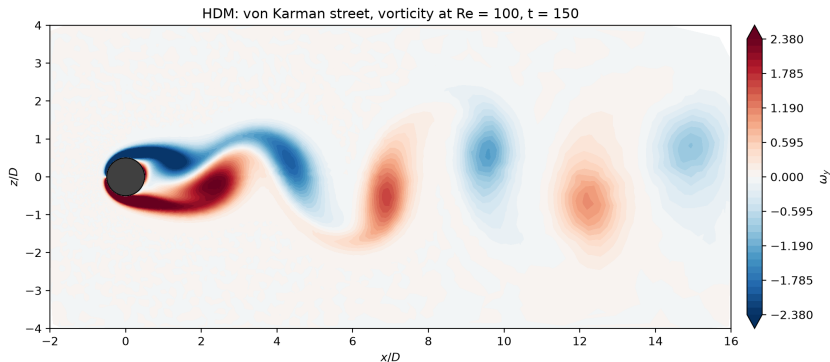
Local Reduced Meshes (ECSW), $N_c = 3$ Local HPROM ($N_e = 3,405$)Local HQPROM ($N_e = 2,152$)Local HPROM-GPR ($N_e = 811$)

Burgers Observations

- ▶ Clustering accelerates all three families, monotonically from global to $N_c = 3$ to $N_c = 5$.
- ▶ The linear basis has the most to gain from it, since it starts without any compression of its own, and by $N_c = 5$ it draws level with local GPR ($132\times$ versus $133\times$).
- ▶ The two mechanisms therefore overlap: clustering and nonlinear compression both reduce the local dimension, so their gains are not simply additive. How much each contributes appears to be problem dependent.

Timings come from a Python implementation and are indicative: the nonlinear models carry the most per-step overhead, so they have the most to gain from vectorization, and their speedups are lower bounds rather than limits of the method. This shows up most in HQPROM against HPROM.

Flow Past a Cylinder Campaign



Global and local HPRM validation on an AERO-F CFD model

Linear POD bases, quadratic manifolds, and ANN, RBF and GPR manifolds

Reproductive regime at $Re = 100$, $M_\infty = 0.2$: one operating point, no parameter dependence

Cylinder Online Performance and Accuracy ($Re = 100$, reproductive)

Computational model	n (N for HDM)	$\bar{n}$	N_e	$\mathbb{W}_{2,C_D}$ (%)	$\mathbb{W}_{2,C_L}$ (%)	Wall-clock time (s)	Speedup
HDM	499 485	–	284 700	–	–	13 000	–
HPROM	35	–	15 791	0.138	0.66	8.00	1 625
HQPROM	10	–	9 443	0.015	0.30	2.67	4 869
HPROM-ANN	5	30	5 459	0.110	1.31	1.88	6 915
HPROM-RBF	5	30	5 143	0.033	0.51	1.40	9 286
HPROM-GPR	5	30	4 911	0.036	0.37	1.33	9 774
Local HPROM ($N_c = 3$)	14–35	–	12 019	0.015	1.09	3.48	3 736
Local HQPROM ($N_c = 3$)	7–10	–	8 645	0.231	4.69	1.83	7 104
Local HPROM-ANN ($N_c = 3$)	3	11–32	3 763	0.208	2.57	1.56	8 333
Local HPROM-RBF ($N_c = 3$)	3	11–32	4 340	0.027	0.81	0.87	14 943
Local HPROM-GPR ($N_c = 3$)	3	11–32	3 395	0.103	1.59	0.71	18 310

Speedup is with respect to $t_{\text{HDM}} = 13\,000$ s; timings are the median of five serial repetitions on an idle machine, 8 MPI ranks (CV 2–4%).

N_e is the number of elements in the ECSW reduced mesh (284 700 for the full mesh).

$\mathbb{W}_2$ is the quadratic Wasserstein distance between the value distributions of the two force histories over the limit cycle, normalized by $\bar{C}_D$ and by C'_L respectively. It measures orbit amplitude and shape and is invariant to phase; frequency accuracy is reported separately in the appendix.

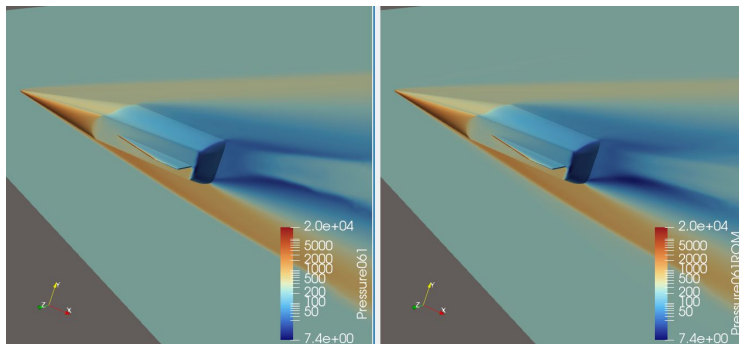
Cylinder Observations

- ▶ The speedup increases monotonically with the degree of nonlinear compression, and in the same order globally and locally: HPROM $\rightarrow$ HQPROM $\rightarrow$ ANN $\rightarrow$ RBF $\rightarrow$ GPR gives $1\,625 \rightarrow 4\,869 \rightarrow 6\,915 \rightarrow 9\,286 \rightarrow 9\,774\times$, and $3\,736 \rightarrow 7\,104 \rightarrow 8\,333 \rightarrow 14\,943 \rightarrow 18\,310\times$.
- ▶ Clustering pays in every family: each local model is faster than its global counterpart, by $1.21\times$ to $2.30\times$, consistent with the Burgers campaign.

Preliminary results: no parameter. One operating point, so the models are tested where they were trained.

Open question: parameterize this case in Re – and then, for the paper, Burgers, the cylinder, or both?

Hypersonic Vehicle Campaign

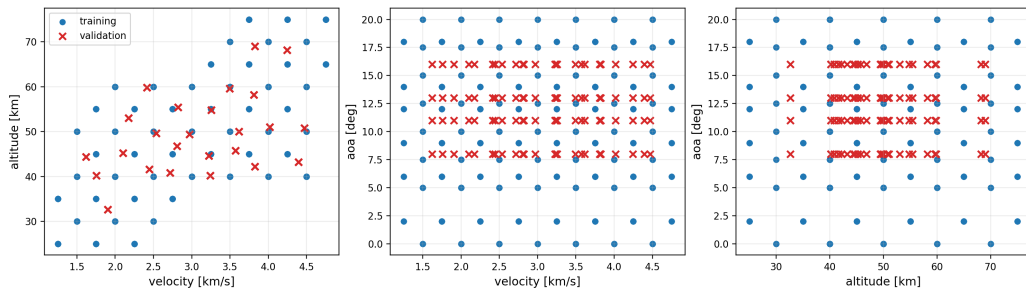


Global and local HPRM validation

Linear POD bases and RBF manifolds

Training and Validation Space

363 HDM training cases; 100 independent HDM validation cases; four local clusters with 20% overlap; ECSW tolerance 10^{-3} .



HGV2 Online Performance and Validation Accuracy

Computational model	n (N for HDM)	$\bar{n}$	C_D (%)	C_L (%)	C_M (%)	Online time (s)	Speedup
HDM	5 674 883	–	–	–	–	<i>pending</i>	–
HPROM	362	–	0.975	0.554	6.620	24.03	–
HPROM–RBF	8	354	1.388	1.335	7.004	1.15	21.0
Local HPROM ($N_c = 4$)	95–168	–	1.101	0.444	6.242	6.80	3.5
Local HPROM–RBF ($N_c = 4$)	8	117–140	1.978	2.183	7.117	0.62	38.9

Coefficient errors are global relative L^2 values over 100 independent validation cases.

Speedup is computed with respect to the HPROM online time, so HPROM itself carries no entry. The HDM runtime is pending.

Timings are means over 10 validation cases, measured on Sherlock using one node, 24 MPI ranks, and 24 cores.

For the HDM, N is the number of mesh nodes. For HPROM, n is the full POD rank. For HPROM–RBF, n and $\bar{n}$ are the primary and secondary dimensions; local $\bar{n}$ is the range across the four cluster bases.

HGV2 Conclusions

- ▶ All four models are equally usable: C_D and C_L within 1–2%, C_M within 6.2–7.1%. The linear models are the most accurate, the RBF models the fastest.
- ▶ RBF closure supplies the compression on its own: $n = 8$ against a full POD rank of 362. Clustering does not reduce n further, but it doubles the speed.
- ▶ The practical gain is where each model runs: the HDM, at $N = 5.7M$, needs Sherlock, while every ROM here answers in under a second on a single node.

HQPROM and Local HQPROM are not included in this campaign.

Supplementary results and implementation details

Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^* = \arg \min_{\mathbf{q}} \frac{1}{2} \|\mathbf{r}(\mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}))\|_2^2.$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta \mathbf{q}$ is obtained from

$$(\Psi^T \mathbf{J}^T \mathbf{J} \Psi) \Delta \mathbf{q} = -\Psi^T \mathbf{J}^T \mathbf{r},$$

where

$$\Psi = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}}, \quad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta \mathbf{q}.$$

Global Linear HPRM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad n = 96.$$

Hence

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \quad \boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Global HQPROM

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}), \quad n = 39.$$

Thus

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}(\mathbf{q} \otimes \mathbf{q})].$$

Global HPROM-GPR

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \quad n = 20, \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}),$$

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}} = \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters $k = 1, \dots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\text{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\text{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\text{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg \min_{\ell=1, \dots, N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \quad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^+ = \arg \min_{\ell=1, \dots, N_c} S_{k^-, \ell}^{\text{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{lin}}(\mathbf{q}) = 2\mathbf{g}_{k^-, \ell}^T \mathbf{q} + d_{k^-, \ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{quad}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{quad}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) + d_{k^-, \ell},$$

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}).$$

HPROM-GPR selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \bar{\mathbf{g}}_{k^-, \ell}^T \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}) + d_{k^-, \ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}_{k^-}), \quad \bar{\mathbf{g}}_{k^-, \ell} := \bar{\mathbf{V}}_{k^-}^T (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$k^+ = k^- \Rightarrow$ no transfer, $k^+ \neq k^- \Rightarrow$ solve transfer problem.

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\text{lin}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+} \mathbf{q}) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\text{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\text{nl}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\text{nl}}(\mathbf{q}) \right\|_2^2.$$

Cluster Change: Nonlinear Least Squares

For nonlinear local manifolds, define

$$\mathbf{r}_{\text{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \quad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}) \Delta \mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\text{tr}}(\mathbf{q}^{(j)}), \quad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta \mathbf{q}.$$

Appendix: Burgers Paper Campaign Setup

Two-dimensional parametric inviscid Burgers problem

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}.$$

$$\mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}, \quad u_x(0, y, t; \boldsymbol{\mu}) = \mu_1.$$

 μ_1 : inflow value μ_2 : exponential source variation $\boldsymbol{\mu} \in [4.25, 5.50] \times [0.015, 0.030]$

Test points:

$$\boldsymbol{\mu}_1 = (4.56, 0.019), \quad \boldsymbol{\mu}_2 = (4.75, 0.020), \quad \boldsymbol{\mu}_3 = (5.19, 0.026).$$

HDM: $\Omega = [0, 100]^2$, 250×250
finite-volume cells.

Methods included:

Global HPROM

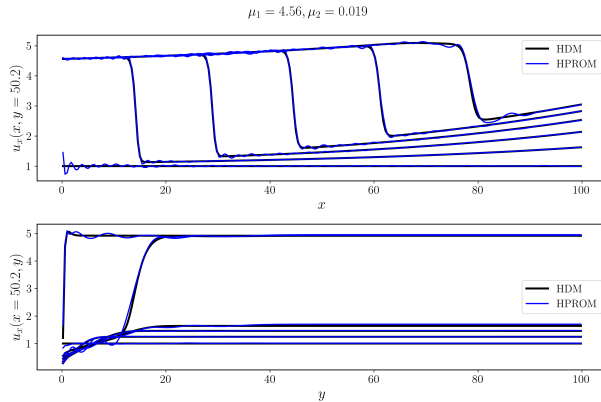
Local HPROM

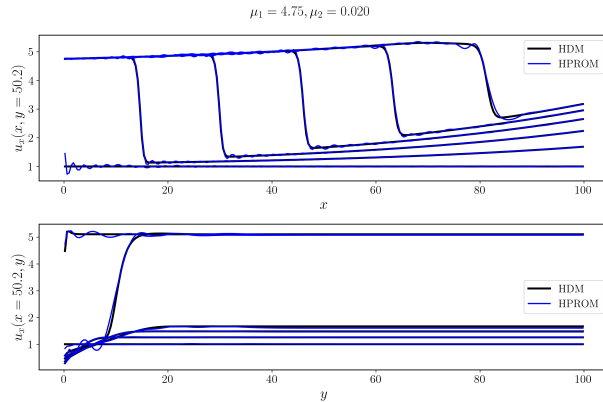
Global HQPROM

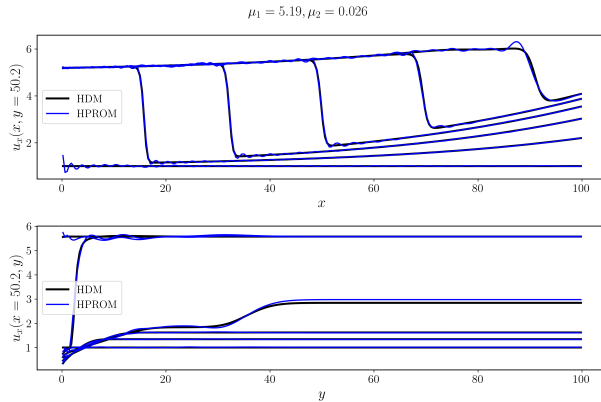
Local HQPROM

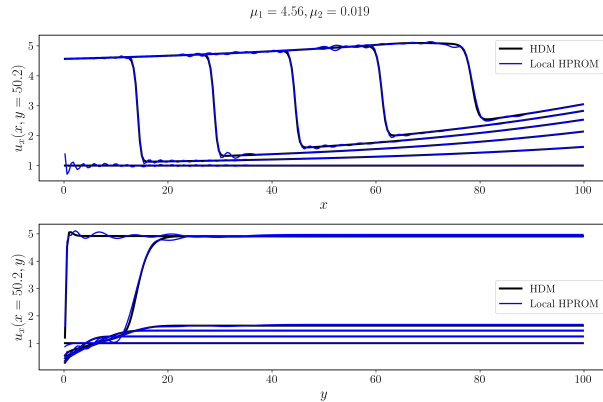
Global HPROM-GPR

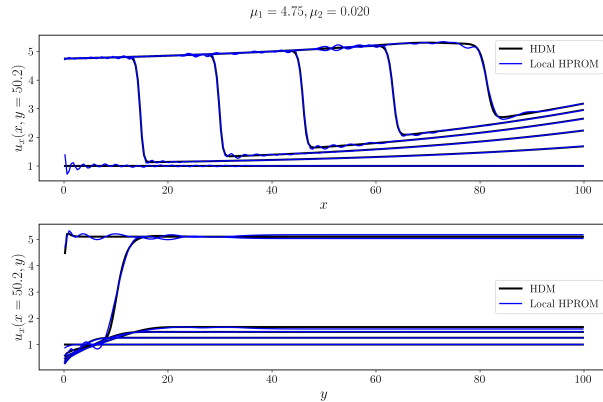
Local HPROM-GPR

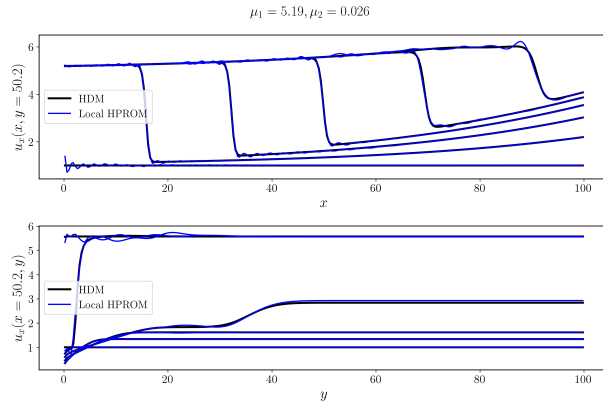
HPROM vs HDM: $\mu = (4.56, 0.019)$ 

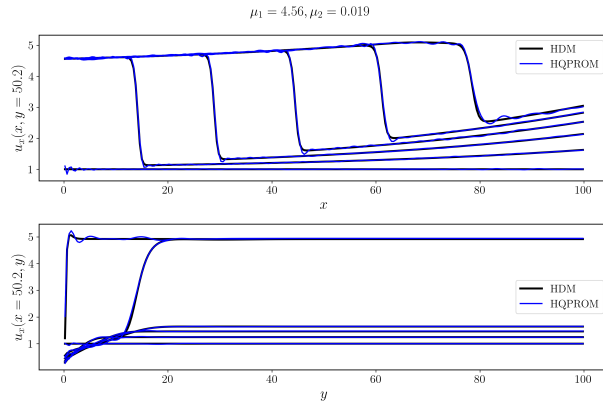
HPROM vs HDM: $\mu = (4.75, 0.020)$ 

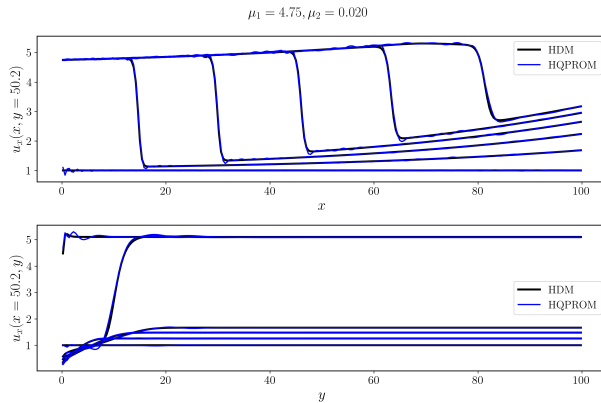
HPROM vs HDM: $\mu = (5.19, 0.026)$ 

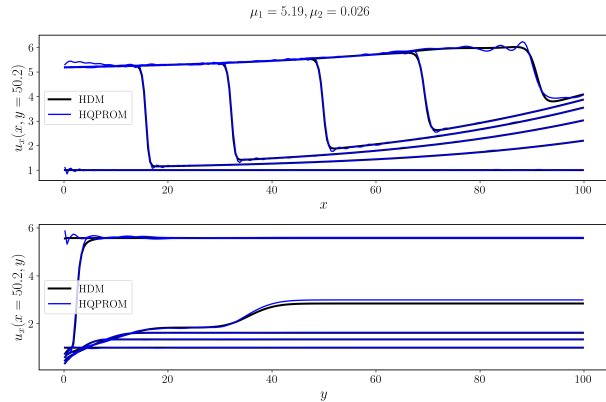
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

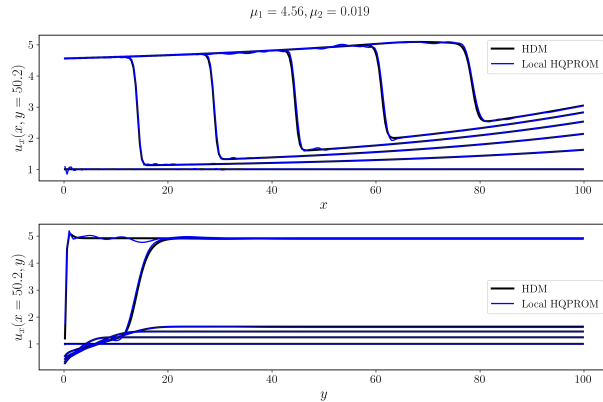
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

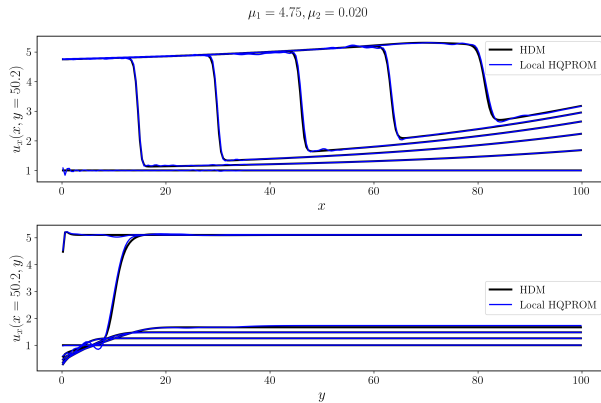
Local HPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

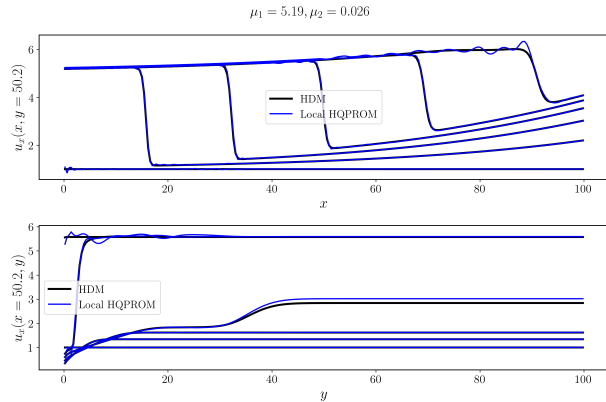
HQPROM vs HDM: $\mu = (4.56, 0.019)$ 

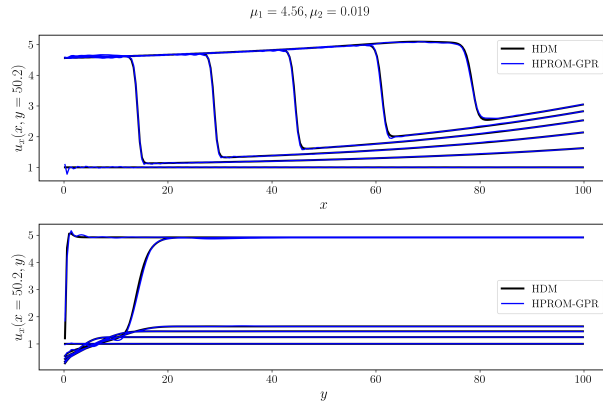
HQPROM vs HDM: $\mu = (4.75, 0.020)$ 

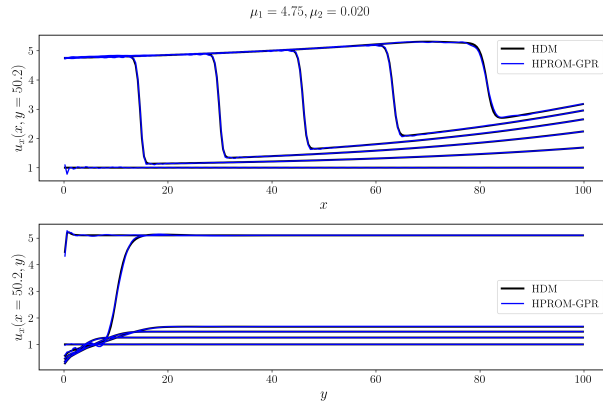
HQPROM vs HDM: $\mu = (5.19, 0.026)$ 

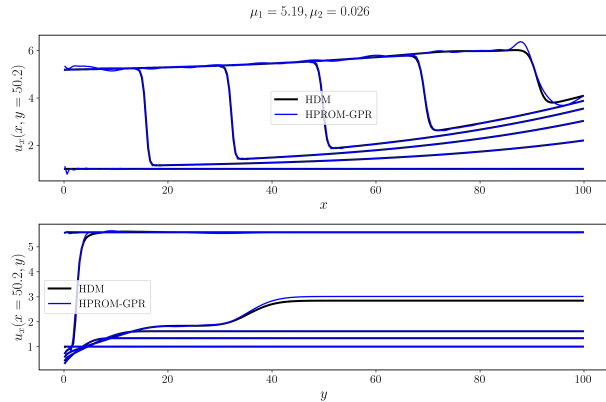
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

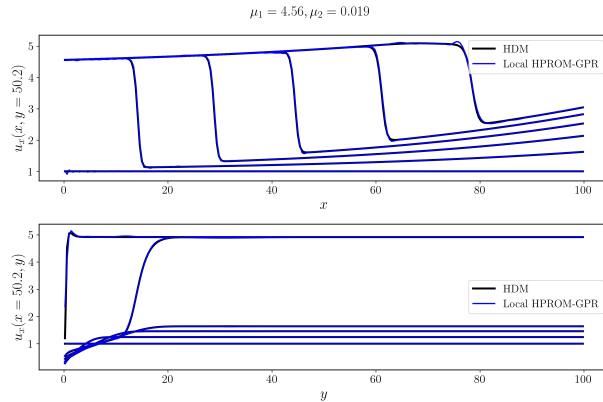
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

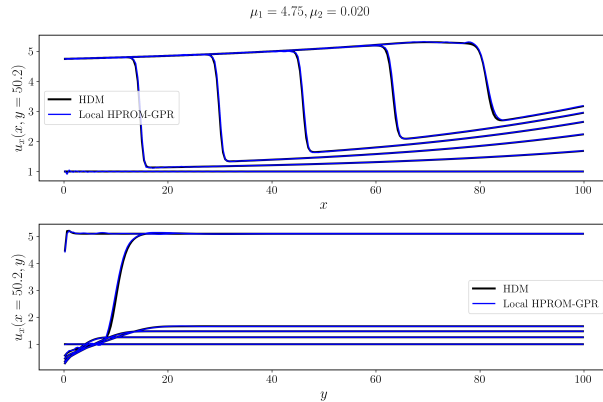
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

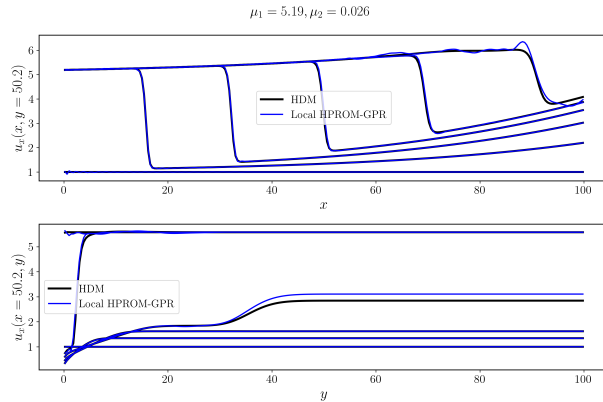
HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$ 

HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$ 

HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

Appendix: Cylinder Governing Equations and Discretization

Compressible Navier–Stokes, perfect gas, laminar (no turbulence closure).

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = \nabla \cdot \mathbf{F}_v(\mathbf{U}, \nabla \mathbf{U}), \quad \mathbf{U} = [\rho, \rho \mathbf{v}, \rho E]^T \in \mathbb{R}^5.$$

Fluid model	γ	1.4	Spatial discretization	
	R	287.1 J/(kg K)	operator	finite volume
	viscosity	constant, $\mu = 10^{-2}$	convective flux	Roe
	turbulence	none (laminar)	reconstruction	linear, $\beta = 1/3$
	L_{ref}	1.0 (cylinder diameter)	limiter	none
Boundary conditions			dissipation	second order
inlet	external, $M_\infty = 0.2$		gradients	least squares
	$\alpha = \beta = 0$		Time integration	
	$\rho_\infty = 1.0, p_\infty = 17.857$		scheme	implicit, RK2
wall	adiabatic, full integration		Δt	0.1, $t \in [0, 150]$
far field	Steger–Warming		Jacobian	exact matrix-vector
			Newton	10 its, $\varepsilon = 10^{-4}$
			linear solver	GMRES(100), $\varepsilon = 10^{-4}$
			preconditioner	RAS, fill 1

No limiter is used because the flow is smooth and subsonic at $M_\infty = 0.2$; the $\beta = 1/3$ linear reconstruction gives the scheme its second-order character. The domain is $[-20, 20]^2$ in units of the diameter, discretized with 98 140 nodes and 284 700 elements, for $N = 499\,485$ degrees of freedom; it is run as a one-cell-thick extrusion with symmetry planes. Snapshots are written every two steps, giving 751 per run. MPI

Appendix: Cylinder ROM and Hyperreduction Settings

Basis construction

snapshots	751 (one operating point)
reference state	startup solution
normalization	on
compression	thin SVD via QR
energy criterion	$1 - \varepsilon_S^2 = 0.9999$
resulting n	35

Clustering (local models)

algorithm	k-means++
N_c	3
max iterations	150
overlap	10%
random seed	42 (pinned)
partition	167 / 298 / 289

Projection and hyperreduction

projection	LSPG, Gauss–Newton
least squares	normal equations
Jacobian	frozen
hyperreduction	ECSW
node selection	sampling/weighting, QR
tolerance	10^{-2}
scaling/centering	off

Quadratic manifold

solver	Tikhonov
regularization	GCV, global
SV sampling frac.	0.6
n	10 (global)
columns of $\mathbf{H}$	$n(n+1)/2 = 55$

Closure maps. ANN: one hidden layer of 1000 ReLU units, internal standardization, Adam with plateau-based decay, early stopping on a 10% validation split, exported as TorchScript. RBF: grid search over Gaussian, inverse-multiquadric, multiquadric and linear kernels and 100 values of $\epsilon \in [0.2, 20]$, min-max scaling, selected on a 10% validation split. GPR: Matérn-3/2 with optimized hyperparameters, four restarts, standardized inputs and outputs. All three are seeded, so the reported errors are reproducible bit for bit.

Appendix: Cylinder Accuracy Breakdown

Computational model	St	St error (%)	C'_L error (%)	$\bar{C}_D$ error (%)	$\mathbb{RE}_{2,C_L}$ (%)
HDM	0.16557	–	–	–	–
HPROM ($n = 35$)	0.16560	+0.02	–0.71	+0.14	1.74
HPROM ($n = 10$)	0.16586	+0.17	–10.68	+0.72	14.98
HQPROM	0.16559	+0.01	+0.09	+0.02	19.07
HPROM–ANN	0.16516	–0.25	–1.39	–0.11	35.78
HPROM–RBF	0.16520	–0.23	–0.27	–0.03	18.99
HPROM–GPR	0.16545	–0.08	–0.22	+0.04	5.64
Local HPROM ($N_c = 3$)	0.16530	–0.16	–0.95	–0.01	5.06
Local HQPROM ($N_c = 3$)	0.16385	–1.04	–4.38	+0.23	69.41
Local HPROM–ANN ($N_c = 3$)	0.16470	–0.53	–2.66	–0.21	53.67
Local HPROM–RBF ($N_c = 3$)	0.16524	–0.20	–0.36	–0.03	9.96
Local HPROM–GPR ($N_c = 3$)	0.16499	–0.35	–1.67	–0.10	38.22

Measured over $t \in [90, 150]$, where the limit cycle is stationary (about 10 shedding cycles); C'_L is the rms lift and $\bar{C}_D$ the mean drag. $\mathbb{RE}_{2,C_L}$ is the time-domain relative L^2 error of the lift over $t \in [0, 150]$, included for reference.

The last column is why the main table reports the spectral norm instead. Every model resolves the shedding frequency to within 1.04% and the mean drag to within 0.72%, yet the time-domain lift error spans 1.7 to 69.4%: a 1% frequency error integrates into tens of percent over the 25 shedding cycles of the window. HQPROM is the clearest case – +0.01% in St , +0.09% in C'_L and +0.02% in $\bar{C}_D$, the most accurate of the set at $n = 10$, yet 19.07% in time-domain lift. The $n = 10$ linear HPROM is the converse: mid-table on $\mathbb{RE}_{2,C_L}$ while carrying a genuine 10.68% amplitude error.

Appendix: HGV2 ROM Configurations

Model	Reduction	Closure	N_c
HPROM	one POD basis	–	1
HPROM–RBF	$n = 8$ primary coordinates	one RBF map	1
Local HPROM	one POD basis per cluster	–	4
Local HPROM–RBF	$n = 8$ primary coordinates	one RBF map per cluster	4

Every POD basis and RBF map is trained from the same 363 HDM parameter cases. Each online state is assigned to its nearest cluster. ECSW is an independent offline construction step.

Appendix: ECSW Training Data

For `DataType = Jacobian`, each selected parameter solution assigned to cluster k supplies m_k^2 candidate ECSW rows, where $m_k = \min(r_k, \text{MaxNumResidualComponents})$ and `MaxNumResidualComponents` = 30. With row stride s , CM reads every s -th row of its `training_rows` sequence.

Model	ECSW cases	Row stride	m_k	Rows read by CM
Global linear HPROM	30	1	$\min(362, 30) = 30$	$30 \times 30^2 = 27,000$
Local linear HPROM, $N_c = 4$	30	1	$\min(r_k, 30) = 30$	$30 \times 30^2 = 27,000$
Global HPROM–RBF, $n = 8$ (ablation)	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1,920$
Global HPROM–RBF, $n = 8$ (final)	363	4	$\min(8, 30) = 8$	$363 \times 8^2/4 = 5,808$
Local HPROM–RBF, $N_c = 4$, $n = 8$	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1,920$

Interpretation

An ECSW snapshot is a Jacobian-training row, not a physical-time state. The RBF primary dimension $n = 8$ —not $\bar{n}$ —sets the 8×8 count. A row stride of 4 reads rows 0, 4, 8, . . . within every selected parameter case; it does not subsample the 363 parameter cases and does not refer to online Newton iterations.

Appendix: HGV2 Reduced-Mesh Construction

Model	L0	L1	L2	L3	Nodes	Elms.	$N_{\neg L2}$
Global linear HPROM	5,089	36,971	166,878	60,623	269,561	563,301	92,683
Local linear HPROM, 4 clusters	3,484	28,145	169,798	47,183	248,610	470,649	78,812
Global HPROM–RBF, $n = 8$	1,715	15,809	173,072	28,648	219,244	350,921	46,172
Local HPROM–RBF, 4 clusters, $n = 8$	1,113	10,516	175,813	20,602	208,044	298,219	32,231

L0: selected ECSW sample nodes.

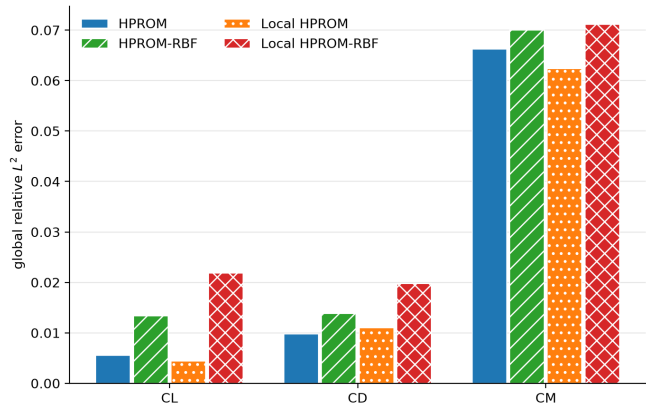
L1: nodes added by edge connectivity for linear reconstruction.

L2: lift-surface nodes used only for C_L , C_D , and C_M postprocessing.

L3: nodes added by element connectivity for state reconstruction.

$N_{\neg L2}$ excludes the postprocessing-only lift-surface nodes.

HGV2 Validation Error Comparison



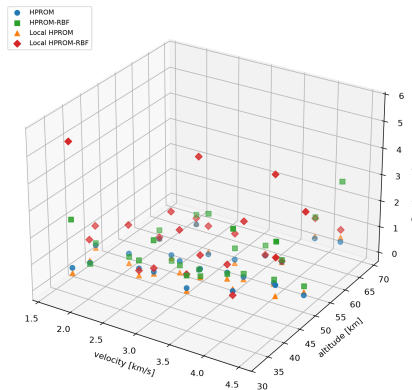
HGV2 ECSW Offline Cost

Computational model	Training rows	Sample nodes	ECSW (h)
HPROM	27 000	5 089	30.04
HPROM–RBF	5 808	1 715	0.43
Local HPROM ($N_c = 4$)	27 000	3 484	9.95
Local HPROM–RBF ($N_c = 4$)	1 920	1 113	0.38

ECSW was run on Sherlock using 20 nodes, 480 MPI ranks, and 480 cores. Times are maximum-rank wall times. Sample nodes is the size of the reduced mesh. How the training rows are assembled is detailed in the appendix.

HGV2 Three-Dimensional C_L Error Field

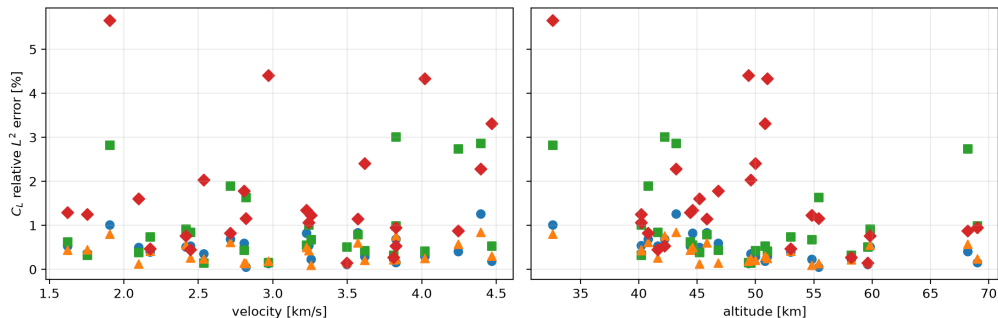
Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_L error over the four sampled angles of attack.



HGV2 Two-Dimensional C_L Error Projections

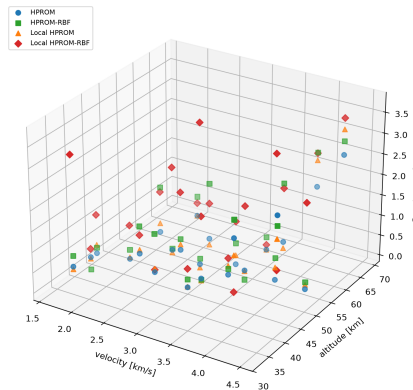
The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.

● HPROM ■ HPROM-RBF ▲ Local HPROM ◆ Local HPROM-RBF



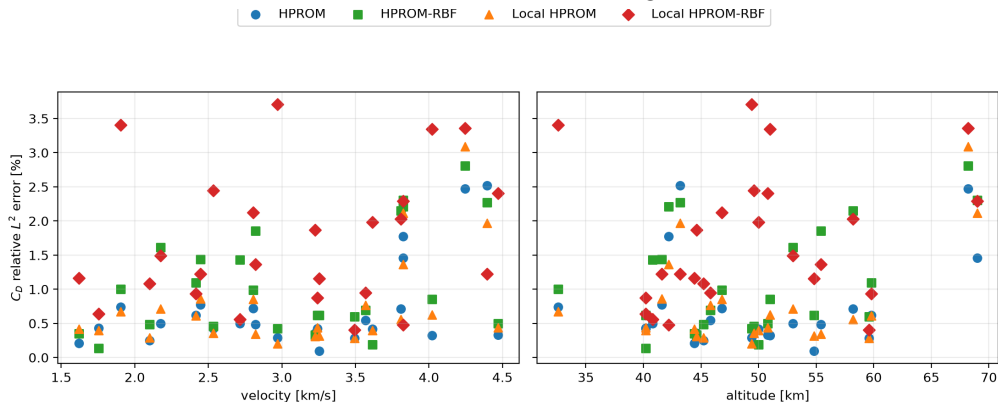
HGV2 Three-Dimensional C_D Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_D error over the four sampled angles of attack.



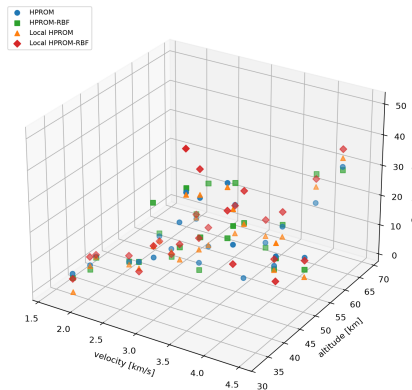
HGV2 Two-Dimensional C_D Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



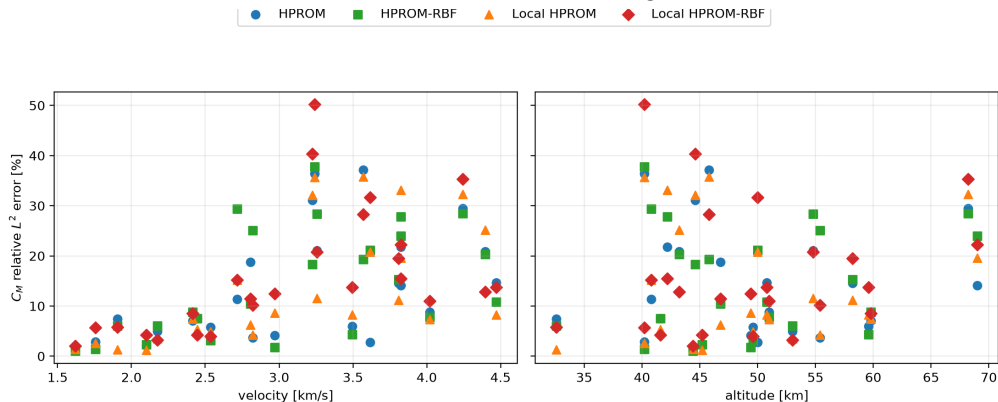
HGV2 Three-Dimensional C_M Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_M error over the four sampled angles of attack.



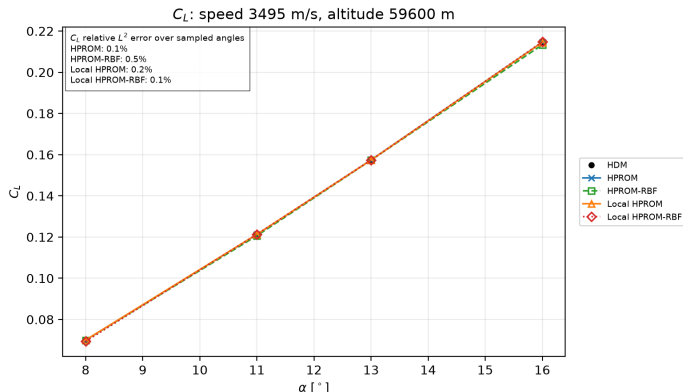
HGV2 Two-Dimensional C_M Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



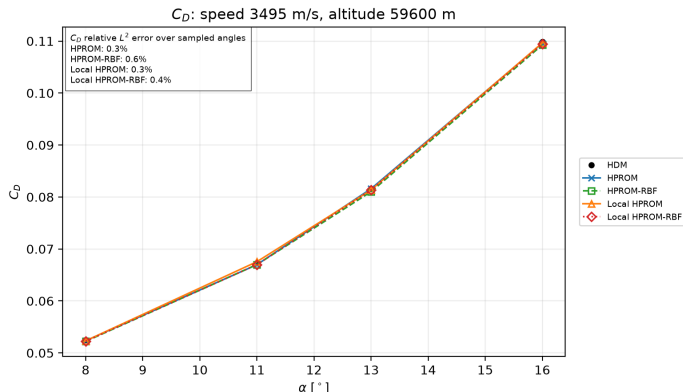
HGV2 Representative Median Pair: C_L

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



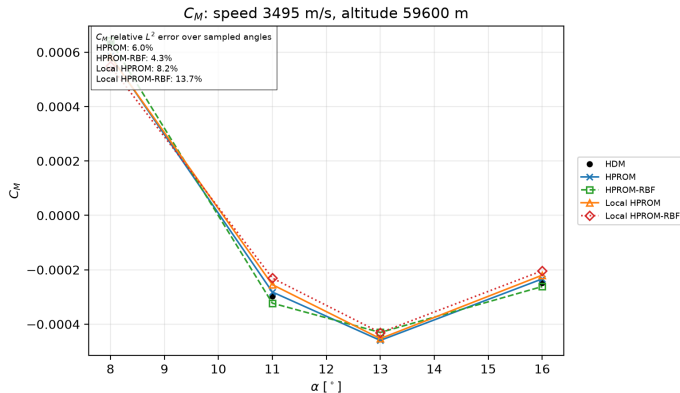
HGV2 Representative Median Pair: C_D

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



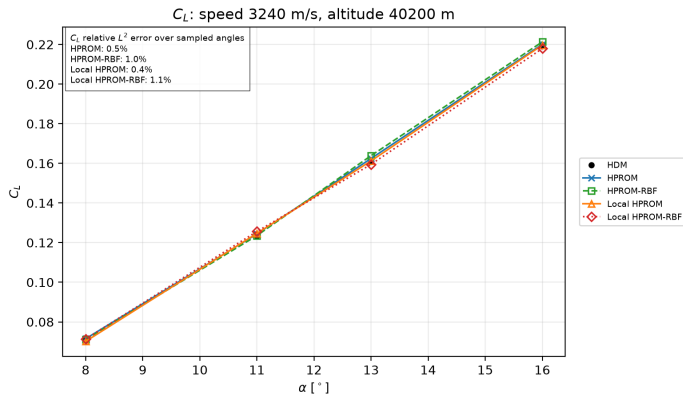
HGV2 Representative Median Pair: C_M

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



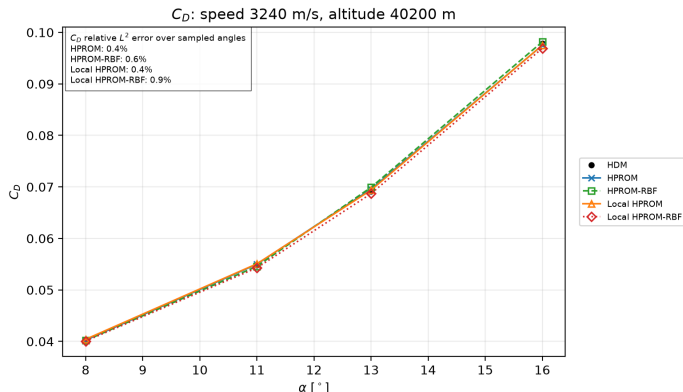
HGV2 Representative Challenging Pair: C_L

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across C_L , C_D , and C_M .



HGV2 Representative Challenging Pair: C_D

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across C_L , C_D , and C_M .



HGV2 Representative Challenging Pair: C_M

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across C_L , C_D , and C_M .

